

Molnár Gate A3 Meeting Summary

Force–displacement h -convergence at $l_c = 0.015$ mm — supervisor decisions
Analysis revision: **db4c1fa** · No additional simulations are requested at this stage

Purpose: (1) accept H2-PUB as the uniform RF–U reference mesh for later work; (2) decide whether matched-state SDV15/crack-path contours are mandatory before the benchmark-reproduction stage, Gate A3, can be accepted.

Definitions used in this summary

Meshes and sizes

- **H0:** original author-supplied supplementary mesh, with local element size $h \approx 0.00494$ mm.
- **H1:** intermediate refined mesh, with $h = 0.0025$ mm.
- **H2-PUB:** finest mesh, generated using the publication-reported local resolution $h = 0.001$ mm. It uses the published resolution, but not necessarily the exact published mesh topology.
- N : number of physical finite elements in the model. It does not include the additional layered visualization/solution elements.
- h : local element size in the expected crack-propagation region.
- l_c : phase-field regularization length scale; here $l_c = 0.015$ mm.

Quantities and stages

- **RF2 and U2:** reaction force and prescribed displacement in the loading direction.
- **NRMSE:** normalized root-mean-square difference between two force–displacement curves.
- **SDV15:** phase-field fracture variable, where 0 represents intact material and 1 represents fully broken material.
- **Pre-peak:** response before the maximum reaction force.
- **Post-peak:** response after the maximum reaction force.
- **Gate A3:** the project’s benchmark-reproduction acceptance gate.
- **MISESeri:** the Abaqus stress-discretization error indicator intended for the later mesh-refinement workflow.

Status

- Solvers H0 / H1 / H2-PUB: technical pass · RF–U packages complete
- Peak / pre-peak h -convergence: **supported**
- Post-peak h -convergence: **not fully demonstrated**
- Crack-path / SDV15 contours: **not assessed**
- Gate A3: **open** — supervisor decision pending

Mesh family

Mesh	Local element size h [mm]	Physical elements N	Description
H0	0.00494	3,930	Original author-supplied supplementary mesh
H1	0.00250	12,064	Intermediate refined mesh
H2-PUB	0.00100	33,852	Fine mesh using the publication-reported local resolution

RF–U successive differences

Pair	ΔF_{peak}	ΔU_{peak}	pre-peak NRMSE	full NRMSE	post-peak NRMSE
H0→H1	4.003%	5.172%	0.387%	21.306%	62.904%
H1→H2-PUB	0.469%	0%	0.106%	6.011%	20.206%

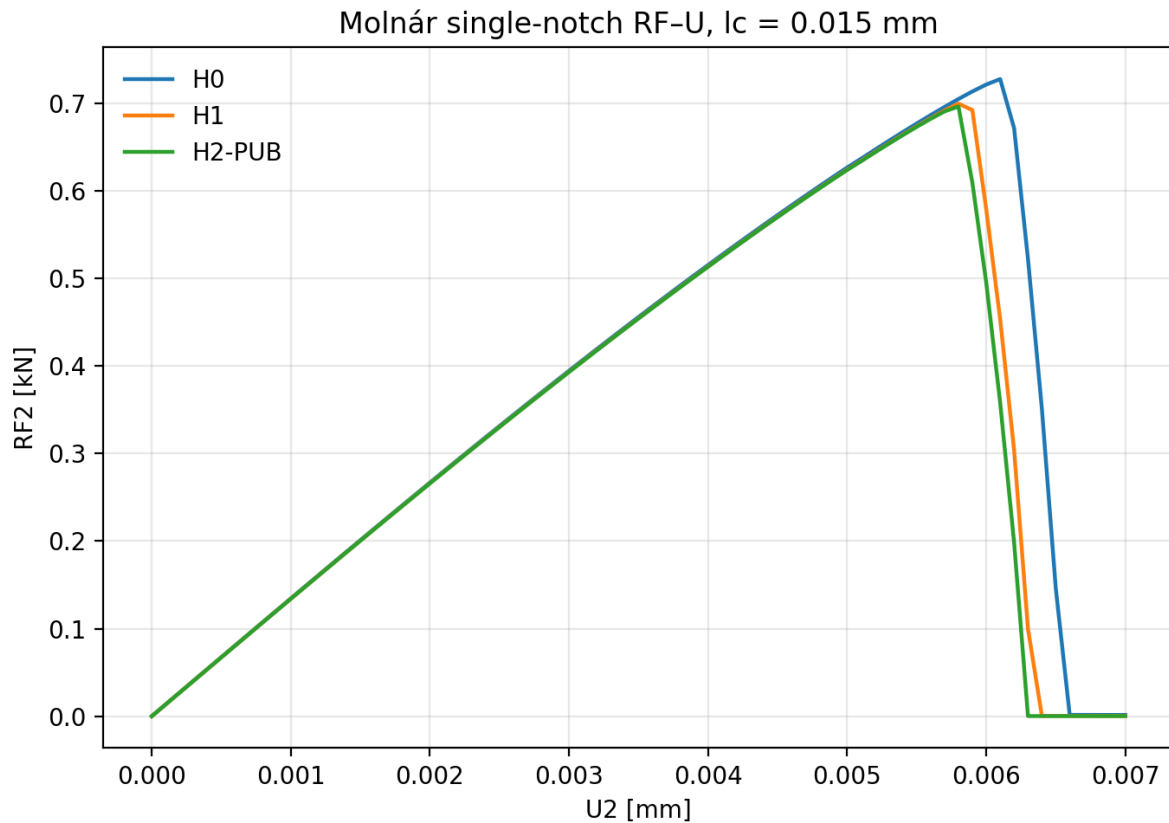
Peak RF2: H0 0.727608 · H1 0.699604 · H2-PUB 0.696336 kN U_{peak} : H0 0.00610 · H1/H2 0.00580 mm

Cost (serial)

Mesh	Walltime	Peak memory (approx.)
H0	00:16:29	~ 0.68 GB
H1	00:46:26	~ 0.91 GB
H2-PUB	02:12:38	~ 1.76 GB

H1 uses about **35%** of H2-PUB walltime with nearly the same peak/pre-peak responses.

RF–U overlay



Decisions requested

1. Uniform RF–U reference

- ☐ **A** Accept H2-PUB as the uniform RF–U reference
- ☐ **B** Accept H2-PUB provisionally (subject to contour evidence)
- ☐ **C** Do not accept; specify additional required evidence

2. Contour requirement

- ☐ **A** Contours mandatory before the benchmark-reproduction stage, Gate A3, can be accepted
- ☐ **B** Contours deferred; RF–U evidence sufficient for conditional preparation of the later MISESeri-based mesh-refinement workflow
- ☐ **C** Contours required only for H1 and H2-PUB
- ☐ **D** Other (please specify)

Recommended conclusion (project)

The intermediate mesh H1 and the fine mesh H2-PUB produce nearly identical elastic, pre-peak and peak-load force–displacement responses. H2-PUB is recommended as the conservative uniform reference because it uses the publication-reported local crack-path resolution of $h = 0.001$ mm. H1 may be used for intermediate development studies at lower computational cost. Noticeable differences remain after peak load, and convergence of the crack path has not yet been assessed.

Boundary

No additional simulations are requested at this stage. No MISESeri / remeshing / state transfer until Decisions 1 and 2. A detailed review is available on request.